

JEE-Main-29-01-2025 (Memory Based)
[EVENING SHIFT]

Physics

Question: An equiconvex lens is cut in two ways as shown. If the focal length of the parts are as mentioned in the diagram. Find $\frac{L_1}{L_2}$.

Options:

- (a) 2
- (b) 4
- (c) $\frac{1}{2}$
- (d) $\frac{1}{4}$

Answer: (c)

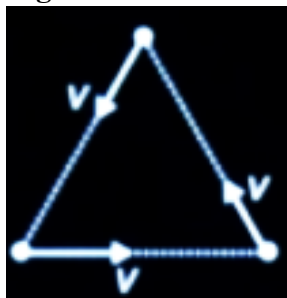
Question: A solenoid of radius 10 cm carrying current 0.29 A and having total 200 turns. If magnetic field inside solenoid is 2.9×10^{-4} T. Find length of solenoid.

Options:

- (a) 6π cm
- (b) 8π cm
- (c) 4.5 cm
- (d) 6 cm

Answer: (b)

Question: Three identical particles, each of mass m move under the influence of mutual attraction forces. Initially they are on the vertices of an equipotential triangle of side 'a' and have equal speed v directed towards the adjacent particles as shown. The net angular momentum about the centre just before collision is



Options:

- (a) $\frac{3mva}{2}$
- (b) $\frac{2}{3}mva$

- (c) $\frac{\sqrt{3}}{2} mva$
- (d) $\frac{2}{\sqrt{3}} mva$

Answer: (c)

Question: Match the physical quantities with their corresponding dimensions

	Column-I		Column-II
(A)	Young's modulus	(i)	$[AL^2]$
(B)	Magnetic moment	(ii)	$[ML^2T^{-2}A^{-1}]$
(C)	Magnetic flux	(iii)	$[AL^{-1}]$
(D)	Magnetic Intensity	(iv)	$[ML^{-1}T^{-2}]$

Options:

- (a) A-(iii), B-(i), C-(ii), D-(iv)
- (b) A-(iv), B-(ii), C-(i), D-(iii)
- (c) A-(iii), B-(i), C-(ii), D-(iv)
- (d) A-(iii), B-(ii), C-(i), D-(iv)

Answer: (a)

Question: Two particles of same mass are performing SHM vertically with two different springs of spring constants K_1 and K_2 . If amplitude of both is same. Find ratio of the maximum speed of two particles.

Options:

- (a) $\sqrt{\frac{K_1}{K_2}}$
- (b) $\sqrt{K_1 K_2}$
- (c) $\sqrt{\frac{K_2}{K_1}}$
- (d) $\sqrt{\frac{K_1 + K_2}{K_1 - K_2}}$

Answer: (c)

Question: A physical quantity Q is given as $Q = \frac{ab^4}{cd}$, if the percentage error is a, b, c and d are 2%, 1%, 2% and 1%, the % error in Q will be

Options:

- (a) 5%
- (b) 15%
- (c) 9%
- (d) 2%

Answer: (c)

Question: Two planets A and B are revolving around a massive star such that $r_A = 2r_B$ and $m_A = 4\sqrt{3} m_B$. Find the ratio of angular momentum of planet B to planet A.

Options:

- (a) $2\sqrt{3}$
- (b) $\frac{1}{4\sqrt{6}}$
- (c) $\frac{1}{2\sqrt{3}}$
- (d) $3\sqrt{2}$

Answer: (b)

Question: Assertion : On increasing the pressure, the volume decrease is more in an isothermal process than in an adiabatic process.

Reason : Adiabatic process is given by PV^γ .

Options:

- (a) Assertion is correct and Reason is false
- (b) Assertion is correct and Reason is correct
- (c) Assertion is false and Reason is correct
- (d) Assertion is false and Reason is false

Answer: (b)

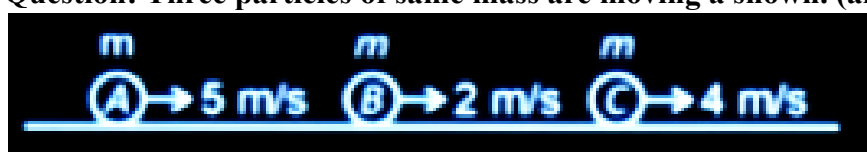
Question: A capacitor $C_1 = 6 \mu\text{F}$, initially charged with a cell of emf 5 V is disconnected and connected to another capacitor $C_2 = 12 \mu\text{F}$ which is initially neutral. The charges on C_1 and C_2 after connection are

Options:

- (a) $0 \mu\text{C}$, $30 \mu\text{C}$
- (b) $10 \mu\text{C}$, $20 \mu\text{C}$
- (c) $20 \mu\text{C}$, $10 \mu\text{C}$
- (d) $30 \mu\text{C}$, $0 \mu\text{C}$

Answer: (b)

Question: Three particles of same mass are moving as shown. (all collisions are elastic)



S_1 : After all collisions velocities are 4 m/s , 2 m/s and 5 m/s .

S_2 : Velocities are interchanged in elastic collisions of the same mass.

Options:

- (a) S_1 : Correct, S_2 : Correct
- (b) S_1 : Incorrect, S_2 : Correct
- (c) S_1 : Incorrect, S_2 : Incorrect
- (d) S_1 : Correct, S_2 : Incorrect

Answer: (b)

Question: An electromagnetic wave propagates in +X-direction. Then, electric field and magnetic field are directed along

Options:

- (a) X, Y
- (b) Y, Z
- (c) Z, Y
- (d) Y, X

Answer: (b)

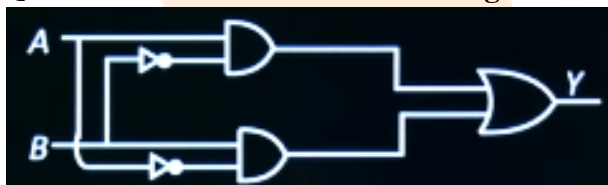
Question: A converging lens of focal length 24 cm, made of glass ($\mu_{\text{glass}} = 1.5$) is immersed completely in water ($\mu_{\text{water}} = 1.33$). It will now behave like a converging lens of focal length _____ cm.

Options:

- (a) 24
- (b) 46
- (c) 96
- (d) 100

Answer: (c)

Question: The truth table for the logical circuit shown below is



Options:

(a)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(b)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

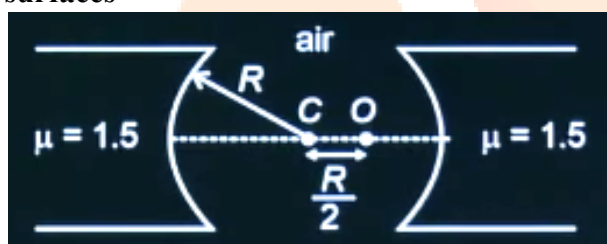
(c)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

(d)

Answer: (b)

Question: Figure shows two spherical surfaces of radius R having a common centre. If the object is placed at O , find the distance between the first images formed by both the surfaces



Options:

(a) $4R/35$

(b) $4R/27$

(c) $4R/70$

(d) $2R/35$

Answer: (a)

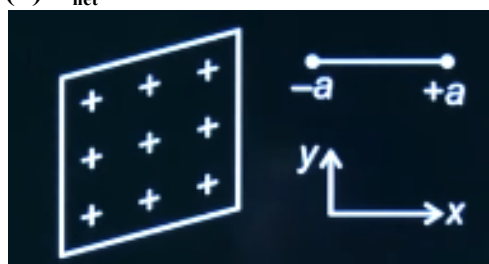
Question: A dipole is placed such that its axis is perpendicular to the infinite charged sheet. Select the correct options

(a) $T_{\text{net}} = 0$, F_{net} is along -ve x-axis

(b) $T_{\text{net}} \neq 0$, $U = \text{min}$

(c) $T_{\text{net}} \neq 0$, $F_{\text{net}} \neq 0$

(d) T_{net} and U both are maximum



Options:

(a) (a), (b), (c) and (d)

- (b) (b) and (c)
- (c) (a) and (c)
- (d) (b) and (d)

Answer: (b)

Question: A cup of coffee takes a time 't' to cool from 90°C to 80°C in a surrounding of 20°C . If a similar cup of coffee is cooled from 80°C to 60°C in the same surrounding, it takes a time

Options:

- (a) $13t/5$
- (b) $5t/13$
- (c) $12t/5$
- (d) $2t$

Answer: (a)

Question: For a certain mechanical system the rate of accretion dm/dt is proportional to $\sqrt{v}$, where m is mass, t time and v is velocity, then the power is proportional to $v^{n/2}$ where n is _____

Options:

- (a) $n = 10$
- (b) $n = 5$
- (c) $n = 15$
- (d) $n = 20$

Answer: (b)

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Chemistry

Question: Which of the following forms the most stable carbocation?**Options:**

- (a) $(\text{Ph})_3\text{C-Br}$
- (b) $\text{C}_6\text{H}_5\text{CH}_2\text{Br}$
- (c) $\text{C}_6\text{H}_5\text{CH}(\text{Br})\text{CH}_3$
- (d) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$

Answer: (a)**Question:** Number of σ and π bonds respectively in hex-1-en-4-yne are**Options:**

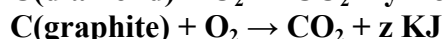
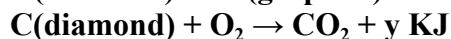
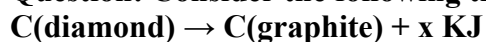
- (a) 13, 3
- (b) 14, 3
- (c) 3, 14
- (d) 14, 13

Answer: (a)**Question:** Which element in group 15 has lowest ionisation Energy**Options:**

- (a) Bi
- (b) P
- (c) As
- (d) Sb

Answer: (a)

Property	Nitrogen	Phosphorus	Arsenic	Antimony	Bismuth
first ionization energy (kJ/mol)	1402	1012	945	831	703
common oxidation state(s)	-3 to +5	+5, +3, -3	+5, +3	+5, +3	+3
ionic radius (pm) [§]	146 (-3), 16 (+3)	212 (-3), 44 (+3)	58 (+3)	76 (+3), 60 (+5)	103 (+3)
electron affinity (kJ/mol)	0	-72	-78	-101	-91

Question: Consider the following thermochemical reactions and choose the correct**Options:**

- (a) $x = y + z$
- (b) $x = y - z$
- (c) $x + y = z$
- (d) $x + y = -z$

Answer: (b)

Question: Which of the following will give azo dye test?

Options:

- (a) Aniline
- (b) Anisole
- (c) Benzene
- (d) Benzaldehyde

Answer: (a)

Question: Which of the following is an essential amino acid?

Options:

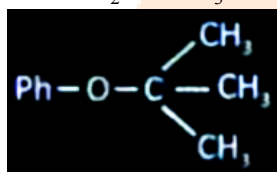
- (a) Alanine
- (b) Glycine
- (c) Valine
- (d) Aspartic acid

Answer: (c)

Question: Which of the following ether react with HBr to form phenol

Options:

- (a) $\text{Ph} - \text{CH}_2 - \text{O} - \text{CH}_2 - \text{CH}_3$
- (b) $\text{Ph} - \text{CH}_2 - \text{OCH}_3$



- (c)
- (d) $\text{Ph} - \text{CH}_2 - \text{O} - \text{CH}_2 - \text{Ph}$

Answer: (c)

Question: A drug becomes ineffective when it decomposes to 50% its concentration. If 16 mg of said drug becomes 4 mg in 12 months, find the time in which drug becomes ineffective given that decomposition of drug follows first order kinetics.

Options:

- (a) 6 months
- (b) 3 months
- (c) 2 months
- (d) 12 months

Answer: (a)

Question: Which of the following gives predominantly O_2 on electrolysis among the following?

- A. Aq. AgNO_3 (Pt electrodes)
- B. Aq. AgNO_3 (Ag electrodes)
- C. Conc. H_2SO_4 (Pt electrodes)
- D. Dilute H_2SO_4 (Pt electrodes)

Options:

- (a) AB
- (b) BC
- (c) ABC
- (d) AD

Answer: (d)

Question: Determine the type of oxide formed by an element (A) which has smallest size among following Li, Na, K, Be, B, Mg

Options:

- (a) A_2O_3
- (b) AO
- (c) AO_2
- (d) A_2O_2

Answer: (a)

Question: Statement-I In partition chromatography a thin film of liquid acts as stationary phase.

Statement-II. Paper chromatography is not a type of partition chromatography.

Options:

- (a) Statement-I is correct and statement-II is incorrect
- (b) Statement-II incorrect and statement-II is correct
- (c) Both statement-I and Statement-II are correct
- (d) Both statement-I and statement-II are incorrect

Answer: (a)

Question: 7.3g Benzalacetone is synthesized from 10.6g of benzaldehyde using acetone as other reactant. Percentage yield of Benzalacetone is

Options:

- (a) 50%
- (b) 27%
- (c) 90%
- (d) 40%

Answer: (a)

Question: Number of spectral lines in the 4th excited state of hydrogen atom

Options:

- (a) 6
- (b) 3
- (c) 10
- (d) 2

Answer: (c)

Question: How many maximum spectral lines are observed when a sample of hydrogen atom d excited from $n = 4$ to n _____

Answer: (6)

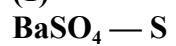
Question: Find spin only magnetic moment of yellow colored complex compound $K_3[Co(NO_2)_6]$, $Cu_2[Fe(CN)_6]$, $Zn_2[Fe(CN)_6]$, $Cu_3[Fe(CN)_6]_2$

Answer: (0)

Question: 0.41 g of $BaSO_4$ is obtained from 0.2 g of organic compound in Carius method. What is the percentage of sulphur present in organic compounds?

Answer: (28)

(1)



$$233\text{g} \text{ — } 32\text{g}$$

$$0.41$$

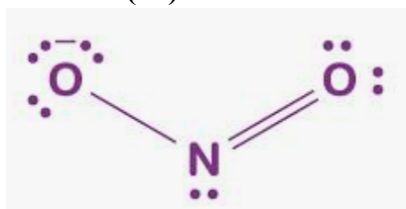
(2)

$$\% \text{ S} = 0.056/0.2 \times 100$$

$$= 28\%$$

Question: Total number of nonbonding electrons in NO_2^-

Answer: (12)



Question: The number of benzenoid structural isomers having molecular formula C_9H_{12} which do not give Baeyer's reagent test is?

Answer: (8)